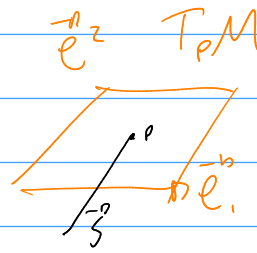
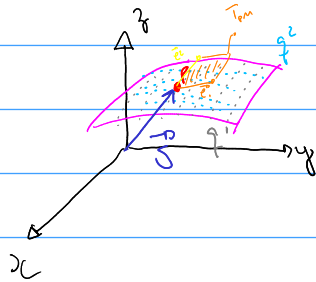


Vetores como operadores:



$$\vec{v} = v^1 \vec{e}_1 + v^2 \vec{e}_2$$

Fibrado Tangente $\rightarrow TM$

$$\vec{e}_i = \frac{\partial \vec{S}}{\partial q^i}$$

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(base natural)

$$\vec{e}_i = \frac{\partial \vec{S}}{\partial q^i}$$

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outras coordenadas

Transformação: $(q^i \rightarrow q^{i'})$

$$\vec{v} = v^i \vec{e}_i = v^{k'} \vec{e}_{k'}$$

$$\text{onde } \vec{e}_{k'} = \frac{\partial \vec{S}}{\partial q^{k'}} = \left(\frac{\partial \vec{S}}{\partial q^i} \right) \frac{\partial q^i}{\partial q^{k'}}$$

$$\vec{e}_{k'} = \vec{e}_i \frac{\partial q^i}{\partial q^{k'}}$$

Logo:

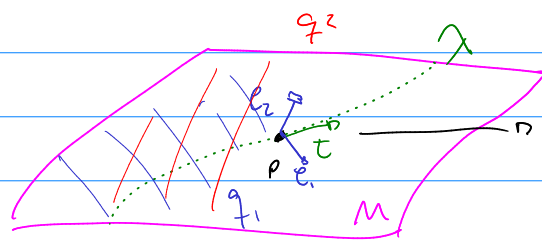
$$\vec{v} = v^i \frac{\partial q^i}{\partial q^{k'}} \vec{e}_i \Rightarrow$$

$$v^{k'} = v^i \frac{\partial q^i}{\partial q^{k'}}$$

Defino: $e_i = \frac{\partial}{\partial q^i}$ no sistema de coord com ... $\rightarrow e_i = \frac{\partial q^k}{\partial q^i} \frac{\partial}{\partial q^k} = \frac{\partial q^k}{\partial q^i} e_k$

Como $\vec{V} = v^i e_i \Rightarrow \vec{V} = v^i \frac{\partial}{\partial q^i}$

Campos Vetoriais: associação suave de $v \in T_p M$ a cada $p \in M$. (C^∞)



$$\dot{\gamma} = \frac{d}{d\lambda} = \dot{\gamma}^i \frac{\partial}{\partial q^i}$$

$$\frac{d}{dq^i} \left(\frac{dq^i}{d\lambda} \right) \Rightarrow \frac{\partial q^i}{\partial \lambda} = \dot{\gamma}^i$$

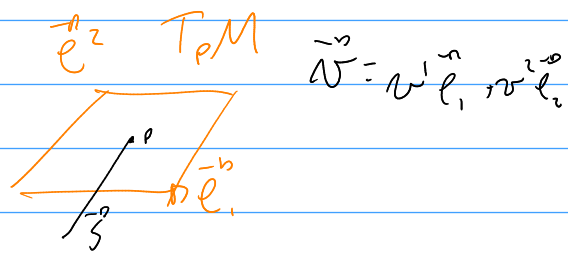
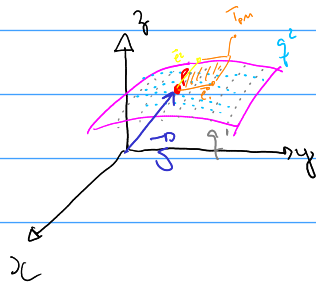
qualquer vetor em $p \in M$ é um vetor tg a alguma curva.

$$\vec{V} = \frac{d}{d\lambda} = \frac{\partial q^i}{\partial \lambda} \frac{\partial}{\partial q^i} = v^i \frac{\partial}{\partial q^i} \text{ onde } v^i = \frac{\partial q^i}{\partial \lambda}$$

Se $\vec{V}$ atua em $q^i(\lambda)$: função coordenada

$$v(q^i(\lambda)) = v^k \frac{\partial q^i(\lambda)}{\partial q^k} = v^k \delta^i_k = v^i$$

Espaço cotangente e 1-formas:



base natural $\rightarrow \vec{e}_i$
 base recíproca $\rightarrow \vec{e}^i$ $\left\{ \vec{v} = v^i \vec{e}_i = v_i \vec{e}^i \text{ onde } \vec{e}_i \cdot \vec{e}^j = \delta_i^j \right.$

onde $v_i = g_{ik} v^k$ e $g_{ik} = \vec{e}_i \cdot \vec{e}_k$

Como $e_i = \frac{\partial}{\partial q^i}$ quem será e^i ?

Sabe-se que:

$$e^{i'} = \frac{\partial q^{i'}}{\partial q^k} e^k$$

e de cálculo elementar $dq^i = \frac{\partial q^i}{\partial q^k} dq^k$

Logo $\rightarrow$ Base dual $\{dq^k\}$ a $\{e_i\}$
 $\hookrightarrow$ espaço cotangente $T_p^* M$

(1-formas ou covetores)

$$\sigma = \sigma_i dq^i \quad \rightsquigarrow \quad d\phi = \frac{\partial \phi}{\partial q^i} dq^i$$

$$\begin{aligned} \langle dq^i, \partial_\kappa \rangle &= \delta^i_\kappa & \langle \sigma, v \rangle &= \langle \sigma_i dq^i, v^\kappa \partial_\kappa \rangle = \sigma_i v^\kappa \langle dq^i, \partial_\kappa \rangle \\ & & &= \sigma_i v^\kappa \delta^i_\kappa \\ & & &= \sigma_i v^i \end{aligned}$$

$$\langle dq^i, \vec{v} \rangle = \langle dq^i, v^\kappa \partial_\kappa \rangle = v^\kappa \delta^i_\kappa = v^i$$

$$\langle \sigma, \partial_i \rangle = \langle \sigma_j dq^j, \partial_i \rangle = \sigma_i$$